

Senthilvel Rajakarthishan

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SUMMARY

As an experienced Data Analyst with expertise in Python, Java, and data visualization tools like Tableau and Power BI, I excel in extracting actionable insights. With a background in cloud-based infrastructure and containerized applications, I have designed data aggregation frameworks, developed pipelines, and automated data processes. My strong problem-solving skills enable me to work efficiently with large datasets, supporting predictive model development and performance testing.

EDUCATION

Master of Science in Information Technology and Management in Data Analytics

Chicago, Illinois

Illinois Institute of Technology

May 2025

- Relevant Coursework: Advanced Topics in Data Management (Oracle Developer)
- Programming for Data analytics (Used R and Python to develop Regression and Classification models)
- Object-Oriented System Analysis, Modeling, and Design.

Bachelor of Engineering in Computer Science

Chennai, India

Sathyabama Institute of Technology

Jun 2023

- Organized a national-level technical symposium "AGON".

TECHNICAL SKILLS

- **Programming & Data Manipulation:** Proficient in Python, R, SQL, and Java, with expertise in data wrangling, ETL, and automation using Pandas, NumPy, and SQLAlchemy.
- **Data Analysis & Visualization:** Skilled in Tableau, Power BI, Excel (Pivot Tables), Matplotlib, Seaborn, and D3.js.
- **Databases & Cloud Technologies:** Strong Knowledge of MySQL, Snowflake, Google BigQuery, AWS (Lambda, RDS), and Azure Synapse for Scalable data management.
- **Machine Learning & AI:** Experience in Scikit-learn, TensorFlow, PyTorch, NLP (NLTK), and OpenCV for predictive modeling and AI applications.

WORK EXPERIENCE

Indium Software Intern

Chennai, India

Data Visualization

May 2021 - Jan 2022

- Solicited data from multiple sources, including databases, spreadsheets, and external providers.
- Acquiring familiarity with fundamental statistical methods and driving a 30% improvement in decision-making process through actionable visualizations and reports.
- Analyzed complex datasets to identify trends and inform marketing strategic insights, led to **20%** increase in click-through rate.
- Explored to various algorithms and performed data analysis using python and visualized data distributions for various features in given customer data.

PROJECTS

Cloud-Based Predictive Maintenance and Anomaly Detection System

- Developed an end-to-end B2B SaaS Sales Intelligence solution leveraging Snowflake, dbt, and Tableau. Built automated ETL pipelines using Python and SQL to ingest CRM (Salesforce) and marketing data.
- Built scalable ETL pipelines and applied anomaly detection models on time-series sensor data from industrial IoT systems.
- Created a real-time dashboard using Power BI to monitor risk scores and performance trends, improving fault detection accuracy by **35%** and reducing downtime.

Healthcare Utilization Dashboard for Public Health Policy Planning

- Created an interactive Tableau dashboard for visualizing healthcare utilization metrics across U.S. states.
- Used SQL and Python to clean and join hospital discharge, population, and cost data.
- Hosted structured data on AWS RDS and built drillable Tableau dashboards showing procedure frequency, stay length, and per capita costs, enabling **state-level resource planning** and reducing manual reporting efforts by **50%**.

HR Employment Data

- Conducted EDA using R and Python on 40,000 records with 25 categories, reducing data dimensionality by **80%** to five key attributes.
- Developed a dynamic linear regression model using subset selection to identify relevant variables.
- Delivered data-driven insights and actionable recommendations to support strategic decision-making.

Sign Language Gesture Recognition using Machine Learning

- Developed a CNN-based real-time gesture recognition model with a **95%** accuracy rate, setting a new industry benchmark.
- Improved model learning efficiency using advanced image preprocessing, normalization, and data augmentation.
- Evaluated performance using accuracy, precision, and recall confirming dependable recognition capabilities and enabled integration for assistive communication tools by optimizing model for deployment on edge devices.